

XML GUIDE FOR DUALSPHYSICS

SHIFTING ADVANCED FORMULATION



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General description

1. The advanced shifting implementation provide a better detection of the free-surface interface.
2. The shifting vector is based on Lind et al., 2012.
3. The free-surface detection method is based on the procedure proposed in Marrone et al. 2016.
4. Arbitrary Lagrangian-Eulerian (ALE) (Oger. et al., 2016) terms can be enabled to improve the volume conservation when shifting is activated.
5. Non-Conservative pressure formulation provide an improve accuracy for the pressure term discretization (Oger et al, 2007, Michel et al., 2023).

Shifting Vector Definition

The procedure is based on the following steps:

1. The normal for particles close to the free-surface is calculated as:

$$\bar{\mathbf{n}}_a = -\frac{\mathbf{L}_a \cdot \nabla C_a}{|\mathbf{L}_a \cdot \nabla C_a|}$$

2. The umbrella region is then scanned and free-surface particles are detected (Marrone et al., 2016)
3. Particles interacting with the free-surface obtained their normal through a smoothed average over free-surface particles:

$$\bar{\mathbf{n}}_a = \frac{\sum_{b \in \Omega_F} \bar{\mathbf{n}}_b W_{ab} V_b}{\sum_{b \in \Omega_F} W_{ab} V_b}.$$

4. The shifting vector is calculated as Lind et al, 2012:

$$\delta \mathbf{x} = -D \nabla C$$

$$D = Ah^2.$$

where the diffusion parameter A is user specified (recommended values between 0.1 and 0.01).

Arbitrary Lagrangian-Eulerian terms

Corrective terms based on an Arbitrary Lagrangian-Eulerian formulation (Oger et al, 2016) are added in the continuity and momentum equation:

$$\frac{d\rho_a}{dt} = \sum_b \left[\frac{\rho_a}{\rho_b} m_b (\mathbf{v}_{ab} + \delta \mathbf{v}_{ab}) \cdot \nabla \mathbf{W}_{ab} + \frac{m_b}{\rho_b} (\rho_a \delta \mathbf{v}_a + \rho_b \delta \mathbf{v}_b) \cdot \nabla \mathbf{W}_{ab} \right] - \mathcal{D}_a,$$

$$\begin{aligned} \frac{d\mathbf{v}_a}{dt} = \sum_b \left[m_b \left(\frac{P_b + P_a}{\rho_a \rho_b} \right) \nabla \mathbf{W}_{ab} + (\mathbf{v}_a \otimes \delta \mathbf{v}_a + \mathbf{v}_b \otimes \delta \mathbf{v}_b) \cdot \nabla \mathbf{W}_{ab} + \right. \\ \left. + \mathbf{v}_a (\delta \mathbf{v}_a - \delta \mathbf{v}_b) \cdot \nabla \mathbf{W}_{ab} \right] + \Gamma_a \end{aligned}$$

$$\frac{d\mathbf{x}_a}{dt} = \mathbf{v}_a + \delta \mathbf{v}_a.$$

Non Conservative pressure formulation

The pressure discretization is modified to increase the accuracy (Oger et al, 2007, Michel et al., 2023). The modified momentum equation is:

$$\begin{aligned} \frac{D\mathbf{v}_a}{Dt} &= -\frac{1}{\rho_a} \sum_b f(p_a, p_b) \nabla_a W_{ab} V_b + \mathbf{f}_a + \mathbf{\Pi}_a \\ f(p_a, p_b) &= \begin{cases} p_a + p_b & \text{if } a \in \Omega_F \cup \Omega_{NF} \\ (p_b - p_a) \mathbf{L}_a & \text{if } a \in \Omega_I \end{cases} \end{aligned}$$

is the correction matrix as defined in Bonet et al, 1999. F and NF refer to free-surface and near free-surface particles, while I refer to internal fluid particles.

XML file: Execution-Parameter: **ShiftingAdv**

- To activate the advanced shifting formulation, select **Shifting=4**.
- The shifting coefficient is defined by the parameter **ShiftAdvCoef**.
- ALE correction terms can be enable/disable with the parameter **ShiftAdvAle=1/0**.
- Non-conservative pressure formulation terms can be enable/disable with the parameter **ShiftAdvNcPress =1/0**

```
<parameter key="Shifting" value="4" comment="Shifting mode 0:None, 1:Ignore bound, 2:Ignore fixed, 3:Full 4:Advanced (default=0)" />
<parameter key="ShiftAdvCoef" value="-0.01" comment="Coefficient for advanced shifting computation (default=-0.01)" />
<parameter key="ShiftAdvALE" value="1" comment="Shifting ALE corrective terms 0:Inactive 1:Active (default=0)" />
<parameter key="ShiftAdvNcPress" value="1" comment="Non-Conservative pressure formulation 0:Inactive 1:Active (default=0)" />
```

ADDITIONAL REMARKS:

- The default shifting coefficient has a different scaling with respect to the standard shifting in DualSPHysics.
- Values recommended are in between -0.1 (flows with strong velocity gradients) to -0.01
- The free-surface shifting threshold, defined by the parameter **ShiftTFS**, is not needed if advanced shifting is enable.
- Viceversa, if **Shifting<4**, **ShiftAdvAle** and **ShiftAdvNcPress** are automatically deactivated

REFERENCES

- Lind, Steven J., et al. "Incompressible smoothed particle hydrodynamics for free-surface flows: A generalised diffusion-based algorithm for stability and validations for impulsive flows and propagating waves." *Journal of Computational Physics* 231.4 (2012): 1499-1523.
- Marrone, Salvatore, et al. "Fast free-surface detection and level-set function definition in SPH solvers." *Journal of Computational Physics* 229.10 (2010): 3652-3663.
- Oger, Guillaume, et al. "SPH accuracy improvement through the combination of a quasi-Lagrangian shifting transport velocity and consistent ALE formalisms." *Journal of Computational Physics* 313 (2016): 76-98.
- Oger, Guillaume, et al. "An improved SPH method: Towards higher order convergence." *Journal of Computational Physics* 225.2 (2007): 1472-1492.
- Michel, J., et al. "A regularized high-order diffusive smoothed particle hydrodynamics scheme without tensile instability." *Physics of Fluids* 35.10 (2023).
- Bonet, Javier, and T-SL Lok. "Variational and momentum preservation aspects of smooth particle hydrodynamic formulations." *Computer Methods in applied mechanics and engineering* 180.1-2 (1999): 97-115.